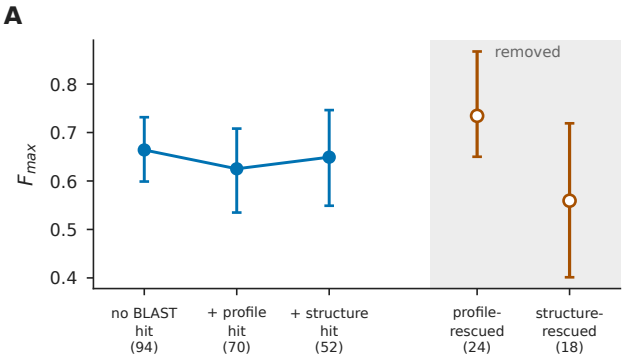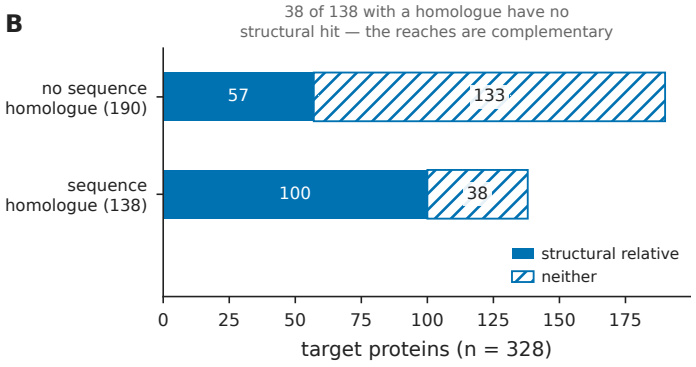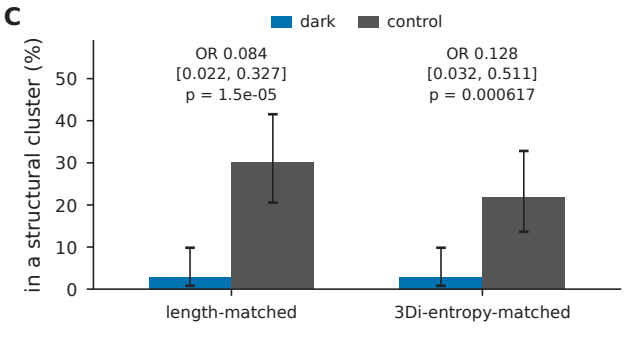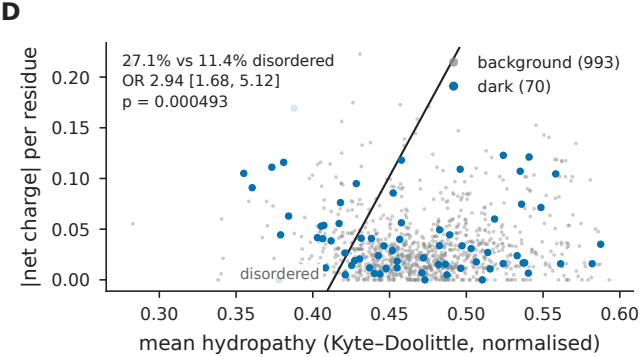**Structural reach and biophysical character of the dark proteome**